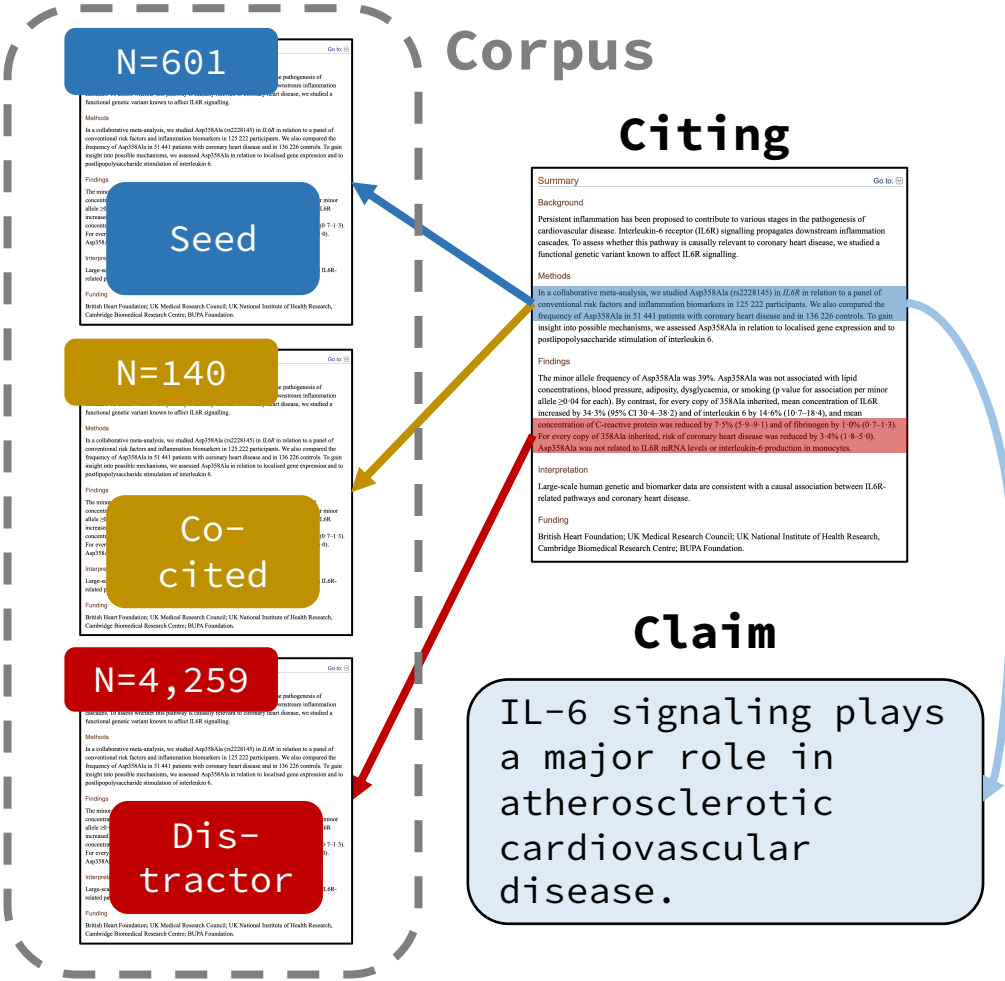
Corpus
N=601
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Summary
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Background
Persistent inflammation has been proposed to contribute to various stages in the pathogenesis of cardiovascular disease. Interleukin-6 receptor (IL6R) signalling propagates downstream inflammation cascades. To assess whether this pathway is causally relevant to coronary heart disease, we studied a functional genetic variant known to affect IL6R signalling.
Methods
In a collaborative meta-analysis, we studied Asp358Ala (rs2228145) in IL6R in relation to a panel of conventional risk factors and inflammation biomarkers in 125 222 participants. We also compared the frequency of Asp358Ala in 51 441 patients with coronary heart disease and in 136 226 controls. To gain insight into possible mechanisms, we assessed Asp358Ala in relation to localised gene expression and to postlipopolysaccharide stimulation of interleukin 6.
Findings
The minor allele frequency of Asp358Ala was 39%. Asp358Ala was not associated with lipid concentrations, blood pressure, adiposity, dysglycaemia, or smoking (p value for association per minor allele ≥0·04 for each). By contrast, for every copy of 358Ala inherited, mean concentration of IL6R increased by 34·3% (95% CI 30·4–38·2) and of interleukin 6 by 14·6% (10·7–18·4), and mean concentration of C-reactive protein was reduced by 7·5% (5·5–9·1) and of fibrinogen by 1·0% (0·7–1·3). For every copy of 358Ala inherited, risk of coronary heart disease was reduced by 3·4% (1·8–5·0). Asp358Ala was not related to IL6R mRNA levels or interleukin-6 production in monocytes.
Interpretation
Large-scale human genetic and biomarker data are consistent with a causal association between IL6R-related pathways and coronary heart disease.
Funding
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Claim
IL-6 signaling plays a major role in atherosclerotic cardiovascular disease.